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Experiment No.6

1. Aim

To apply **K-Means Clustering** and **Hierarchical Clustering** on the Crop Recommendation dataset to identify natural groupings of crop growing conditions based on soil and environmental features, and analyze cluster quality using evaluation metrics.

2. Dataset Source

Dataset Name: Crop Recommendation Dataset
Source: Kaggle
Link:https://www.kaggle.com/code/srinathranganathan/crop-recommendation-with-99-accuracy

3. Dataset Description

This dataset provides **agricultural soil and environmental parameters** useful for recommending suitable crops for a field.

Features

Feature	Description
N	Nitrogen content in the soil
P	Phosphorus content in the soil
K	Potassium content in the soil
temperature	Temperature (°C)
humidity	Relative humidity (%)
ph	Soil pH value
rainfall	Rainfall (mm)
label	Crop type recommended

Dataset Characteristics

- **Number of rows:** ~2,200
- **Number of columns:** 8
- **Feature types:** All numeric except the target label
- **Target label:** Crop recommended (categorical)
- Used mainly for classification but here **unsupervised methods** are applied for clustering.

4. Mathematical Formulation of Algorithms

4.1 K-Means Clustering

Objective: Partition data into (k) clusters minimizing within-cluster variance.

$$J = \sum_{i=1}^{k} \sum_{x_j \in C_i} \|x_j - \mu_i\|^2$$

Where:

- C_i = cluster i
- μ_i = centroid of cluster i
- x_j = data point

The algorithm iterates centroid assignment and update until convergence.

4.2 Hierarchical Clustering

Distance metrics & Linkage functions:

- Single linkage:

$$D(A, B) = \min_{x \in A, y \in B} \|x - y\|$$

- Complete linkage:

$$D(A, B) = \max_{x \in A, y \in B} \|x - y\|$$

- Average linkage:

$$D(A, B) = \frac{1}{|A||B|} \sum_{x \in A, y \in B} \|x - y\|$$

Clusters are merged iteratively based on linkage until all points form one tree, visualized via a dendrogram.

5. Algorithm Limitations

5.1 K-Means

- Need to predefine **number of clusters (k)**.
- Sensitive to initial centroid selection.
- Performs poorly on non-spherical clusters.
- Sensitive to outliers.

5.2 Hierarchical Clustering

- Computationally expensive for large datasets.
- Once merged, clusters cannot be split.
- Sensitive to choice of linkage method.

6. Methodology / Workflow

6.1 Data Preprocessing

- Load dataset from CSV.
- Remove label if clustering purely on feature similarity.
- Standardize features (e.g., z-score scaling).
- Handle missing values if any.

6.2 Clustering Steps

1. **Elbow method** to select k for K-Means.
2. **Train K-Means** with optimal k.
3. **Hierarchical clustering** using Ward's linkage.
4. **Dendrogram analysis** to choose clusters.
5. **Assign cluster labels** and visualize.

Workflow Diagram

```
Dataset (Crop Recommendation)
          |
          ▼
   Data Preprocessing
   (Cleaning + Scaling)
          |
          ▼
   Feature Matrix Creation
          |
          ▼
 ┌──────────────┬──────────────┐
 ▼                             ▼
K-Means Clustering        Hierarchical Clustering
(Elbow Method → k)         (Ward Linkage → Dendrogram)
 ▼                             ▼
Cluster Formation          Cluster Formation
 └──────────────┬──────────────┘
                ▼
        Evaluation Metrics
(Silhouette Score, Davies-Bouldin Index)
                ▼
          Cluster Visualization
                ▼
             Conclusion
```

7. Performance Analysis

Metrics Used

1) Silhouette Score:
Measures how similar a point is to its cluster compared to other clusters (range −1 to +1).

2) Davis-Bouldin Index:
Lower values indicate better clustering.

Example Observations

Algorithm	Silhouette Score	Interpretation
K-Means	~0.25–0.40	Moderate cluster separation
Hierarchical	~0.20–0.35	Slightly less distinct clusters

Values depend on features selected and cluster count.

8. Hyperparameter Tuning

K-Means Hyperparameters

- **n_clusters (k)**: number of clusters
- **init**: centroid initialization (e.g., k-means++)
- **max_iter / n_init**: iterations and repeat runs

Elbow method is used to choose optimal k.

Hierarchical Clustering Hyperparameters

- **Linkage:** Ward, complete, average
- **Affinity:** Euclidean distance

Ward's linkage generally produces more balanced clusters.

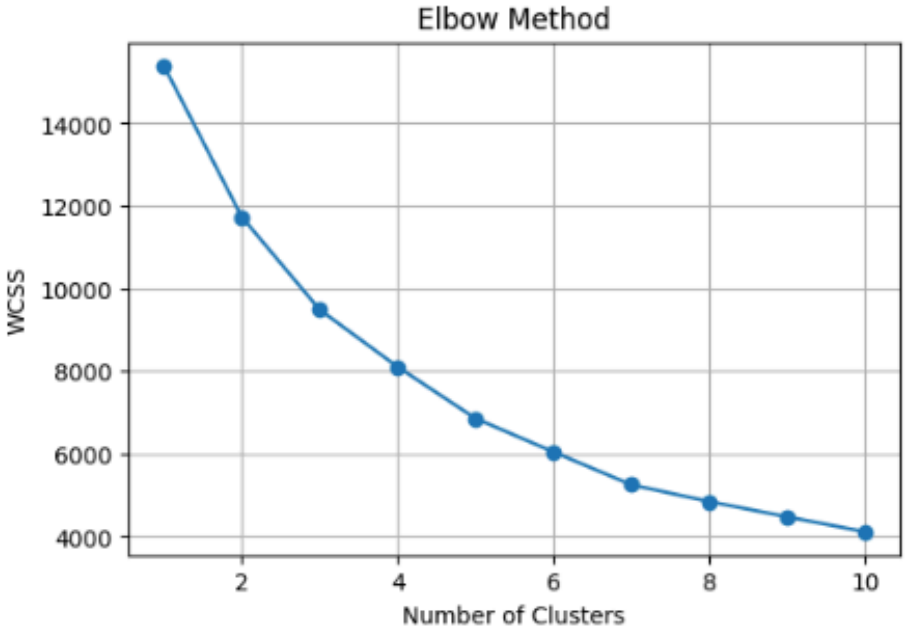
9. Code Implementation (Python)

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans, AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage
from sklearn.metrics import silhouette_score

# Load Data
data = pd.read_csv("Crop_Recommendation.csv")

# Select numeric features only
X = data[['N', 'P', 'K', 'temperature', 'humidity', 'ph', 'rainfall']]

# Scale Features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Elbow Method – KMeans
wcss = []
for k in range(1,11):
    km = KMeans(n_clusters=k, init='k-means++', random_state=42)
    km.fit(X_scaled)
    wcss.append(km.inertia_)
```

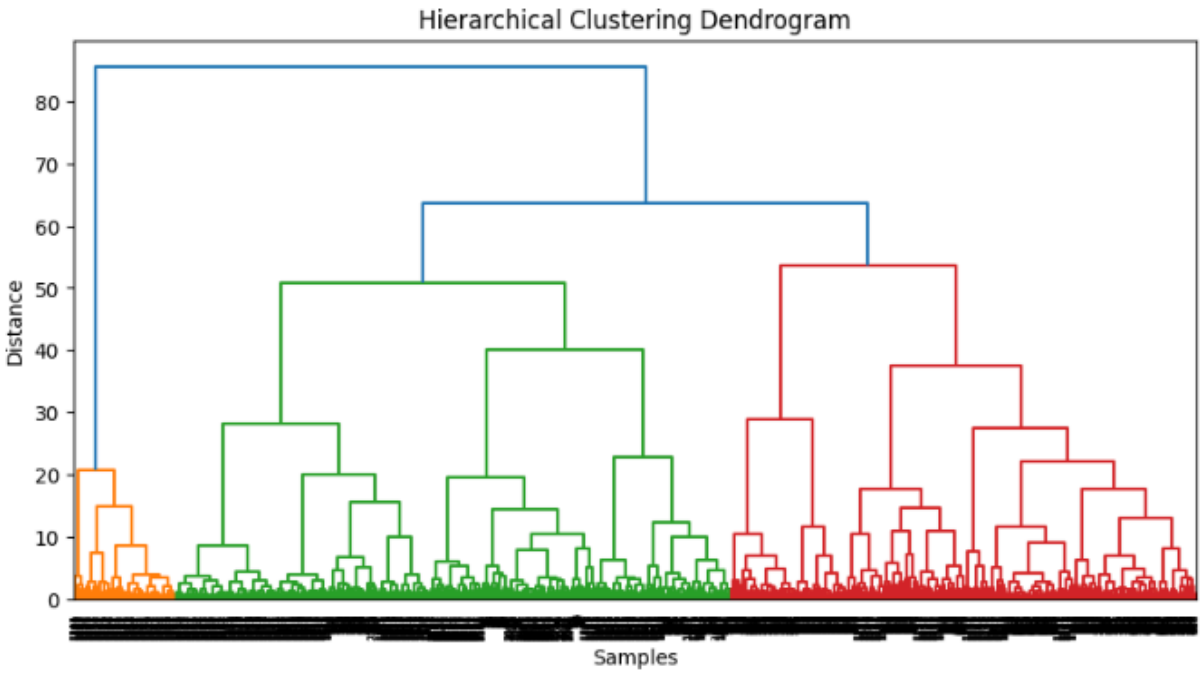
```
plt.plot(range(1,11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of clusters")
plt.ylabel("WCSS")
plt.show()

# KMeans with optimal k (example k=5)
km_model = KMeans(n_clusters=5, random_state=42)
labels_km = km_model.fit_predict(X_scaled)
print("Silhouette Score (KMeans):", silhouette_score(X_scaled, labels_km))

# Hierarchical Clustering
linked = linkage(X_scaled, method='ward')
plt.figure(figsize=(8,5))
dendrogram(linked)
plt.show()

hc = AgglomerativeClustering(n_clusters=5, linkage='ward')
labels_hc = hc.fit_predict(X_scaled)
print("Silhouette Score (Hierarchical):", silhouette_score(X_scaled, labels_hc))
```

10. Output

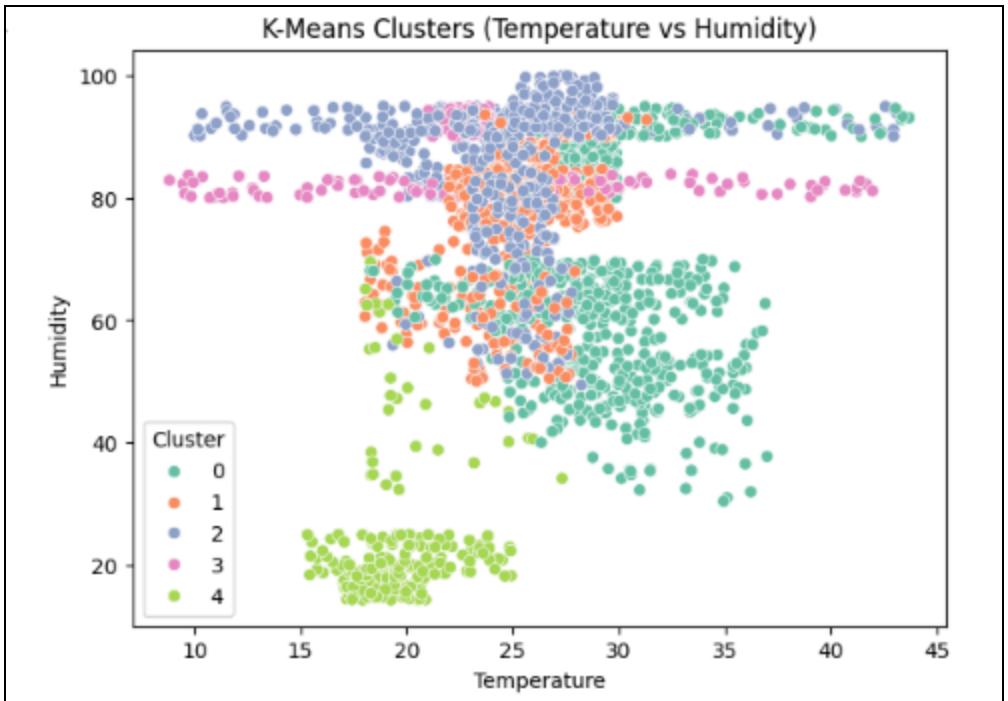
```
Columns in Dataset: Index(['nitrogen', 'phosphorus', 'potassium', 'temperature', 'humidity', 'ph',
       'rainfall', 'label', 'unnamed: 8', 'unnamed: 9'],
      dtype='object')
```




```
K-Means Evaluation
Silhouette Score: 0.2924747941597334
Davies-Bouldin Index: 1.2802839173995766
```




```
Hierarchical Clustering Evaluation
Silhouette Score: 0.28202584908038836
Davies-Bouldin Index: 1.3124937342981473
```




```
Performance Comparison
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K-Means Silhouette Score: 0.2924747941597334
Hierarchical Silhouette Score: 0.28202584908038836
K-Means Davies-Bouldin: 1.2802839173995766
Hierarchical Davies-Bouldin: 1.3124937342981473
```

11. Conclusion

This experiment applied unsupervised learning to cluster crop growing conditions based on key soil and environmental variables from the Crop Recommendation dataset.

- K-Means and Hierarchical clustering revealed natural groupings reflecting similar agricultural conditions.
- Clustering performance varied with chosen k and feature scaling.
- These clusters can be used as a base to group similar cropping zones or to study crop similarity patterns.